

HW 8

- ① - True the waveform is even
- ② - False the duty cycle is 40% so even harmonics are not 0
- ③ - True Discontinuous waveform $\Rightarrow$ infinite
- ④ - True as previous
- ⑤ - False $a_0 = \frac{1 \cdot 2 + 0 \cdot 3}{5} = 0,4$
- ⑥ - False $DC = \frac{2}{5} = 40\%$
- ⑦ - False $f_0 = \frac{1}{T} = \frac{1}{5} = 200 \text{ Hz}$
- ⑧ - True $k=5 \Rightarrow \omega = 2\pi \cdot 5 \cdot 200 = 2000\pi$
- ⑨ - True Infinite period $\Leftrightarrow$ non-periodic signal.

⑩ $a_0 = \frac{A_T}{T}$ $a_0 = 1$, $T = 20 \text{ ms}$, $T = 100 \text{ ms}$

$A = \frac{a_0 T}{T} = \frac{1 \cdot 100}{20} = 5$ 5

⑪ $\frac{T}{T} = \frac{20}{100} = 20\%$

12) 20 ms

13) $f_0 = \frac{1}{T} = \frac{1}{0,1} = 10 \text{ Hz}$

14) $k = 5 \Rightarrow f = 5 \cdot 10 \text{ Hz} = 50 \text{ Hz}$

15) $f_0 = \frac{1}{T} = \frac{1}{0,2} = 5 \text{ Hz}$

16) $k = \frac{T}{\frac{T}{100}} = \frac{200}{100} = 2 \Rightarrow f = 2 \cdot 5 = 10 \text{ Hz}$

17) $f_0 = 5 \text{ Hz}$

18) Average $= -2 \frac{T}{T} = -2 \cdot 0,5 = -1$

19) $a_k = -\frac{2}{k\pi} \sin\left(\frac{k\pi}{2}\right)$ [1]

20) $a_0 = \text{Avg}$ [1]

21) $a_1 = -\frac{2}{\pi} \sin\left(\frac{\pi}{2}\right) = -\frac{2}{\pi}$ [2]

22) $a_2 = -\frac{2}{2\pi} \sin(\pi) = 0$ [6]

23) $a_3 = -\frac{2}{3\pi} \sin\left(\frac{3\pi}{2}\right) = \frac{2}{3\pi}$ [5]

24) $a_4 = -\frac{2}{4\pi} \sin(2\pi) = 0$ [1]

25) $a_6 = -\frac{2}{6\pi} \sin(3\pi) = 0$ [3]